AZURE COMPREHENSIVE STUDY NOTES

AZ-900 Fundamentals + AZ-104 Administrator | Deduplicated & Interview-Ready

STUDY PLAN — HOW THESE NOTES ARE ORGANISED

These notes combine both AZ-900 and AZ-104 course transcripts. Duplicate topics appear once, with AZ-104 depth applied. Each phase below builds on the previous one — follow the order for maximum retention.

- **Phase 1** — Cloud Concepts & Mental Models (AZ-900 foundation — why cloud, service models, cloud types)
- **Phase 2** — Compute Services (VMs, VMSS, Web Apps, Functions, Containers, ACI, ACA, AVD, Batch)
- **Phase 3** — Networking Services (VNets, NSG, ASG, Bastion, Load Balancer, DNS, Traffic Manager, Front Door, Firewall, DDoS, VPN, ExpressRoute, UDR, Network Watcher)
- **Phase 4** — Storage Services (Storage Accounts, Blob, Files, Queue, Table, Data Box, Access Control, Lifecycle, Redundancy)
- **Phase 5** — Database Services — AZ-900 only (SQL DB, MySQL, Cosmos DB, Synapse, Databricks)
- **Phase 6** — Identity & Access (Entra ID, RBAC, Custom Roles, SSPR, MFA, Conditional Access, Domain Services)
- **Phase 7** — Infrastructure as Code (Azure CLI, PowerShell, ARM Templates, Bicep, Modules)
- **Phase 8** — Governance (Policy, Tags, Locks, Management Groups, Resource Moves, Subscriptions)
- **Phase 9** — Monitoring & Backup (Azure Monitor, Log Analytics, Azure Backup, ASR, Backup Vault)
- **Phase 10** — Security Services — AZ-900 (Defender for Cloud, Sentinel, Key Vault, DDoS, Firewall, Zero Trust)
- **Phase 11** — Other Azure Services — AZ-900 (DevOps, Logic Apps, Machine Learning, AI Services, Cost Management)
- **Phase 12** — Interview Rapid-Fire & Comparison Tables

PHASE 1 — CLOUD CONCEPTS & MENTAL MODELS [AZ-900]

1.1 What Is Cloud Computing?

Cloud computing is the delivery of compute services — servers, storage, databases, networking, software, analytics — over the internet. You access these services on demand and pay only for what you use, instead of purchasing and maintaining physical hardware.

⚠ NOTE: You are renting infrastructure from Microsoft's global datacentres instead of owning it. This shifts your spend from CapEx (Capital Expenditure) to OpEx (Operational Expenditure).

Key Benefits

- **Cost** — No upfront hardware cost. Pay only for what you consume — consumption-based model
- **Speed** — Provision infrastructure in minutes, not weeks. No procurement, no shipping, no rack installation
- **Scale** — Scale up or out instantly based on demand
- **Global Reach** — Deploy to Azure regions worldwide — close to your users
- **Security** — Microsoft invests billions in security; physical security of datacentres is their responsibility

1.2 Cloud Service Models

The three service models define the division of management responsibility between the cloud provider (Microsoft) and the customer (you).

Layer	IaaS	PaaS	SaaS
Application	You manage	You manage	Provider manages
Runtime / Middleware	You manage	Provider manages	Provider manages
Operating System	You manage	Provider manages	Provider manages
Virtualisation	Provider manages	Provider manages	Provider manages
Physical Hardware	Provider manages	Provider manages	Provider manages
Azure Examples	VM, VMSS, ACI	App Service, Azure SQL, AKS	Microsoft 365, Dynamics 365

⚠ NOTE: Interview question: 'You need to run a .NET app — which model?' Answer: PaaS (Azure Web App) if you don't want to manage OS. IaaS (VM) if you need OS-level control.

1.3 Cloud Deployment Models

- **Public Cloud** — All infrastructure owned by the cloud provider and shared across customers. E.g., Azure, AWS, GCP. Most cost-effective for most workloads.
- **Private Cloud** — Cloud infrastructure operated solely for one organisation — either on-premises or hosted. Complete control, but full management responsibility.
- **Hybrid Cloud** — Combination of public and private — on-premises datacentre connected to Azure. Allows gradual migration, compliance requirements, and data residency control.

1.4 Shared Responsibility Model

Microsoft is always responsible for the physical datacentre, networking, and virtualisation infrastructure. Your responsibility increases as you move from SaaS → PaaS → IaaS → On-premises.

For IaaS (VMs): Microsoft secures the physical host. You secure the OS, applications, data, and network config.

For PaaS (App Service, SQL): Microsoft secures the OS and runtime. You secure the application code and data.

⚠ **NOTE: Data protection and classification is ALWAYS the customer's responsibility regardless of service model.**

1.5 Cloud Benefits — Deep Definitions

- **High Availability (HA)** — Ensuring an application runs with minimal downtime. In Azure: achieved via Availability Sets, Availability Zones, and redundant deployments.
- **Scalability** — Ability to increase or decrease resources based on demand. Vertical = resize the VM. Horizontal = add/remove VM instances (VMSS).
- **Elasticity** — Automatically scaling resources up and down in real-time in response to actual workload. VMSS with autoscale = elasticity in Azure.
- **Fault Tolerance** — System continues operating even when one or more components fail. Availability Sets spread VMs across fault domains to tolerate hardware failures.
- **Disaster Recovery** — Ability to recover from region-level failures. In Azure: Azure Site Recovery replicates VMs to secondary region; Azure Backup provides restore points.
- **Economies of Scale** — Microsoft's massive purchasing power means per-unit costs are far lower than any individual company could achieve building their own datacentre.

PHASE 2 — COMPUTE SERVICES

2.1 Azure Virtual Machines

An Azure VM is an on-demand IaaS compute resource that runs in Azure's datacentres. You choose the operating system (Windows or Linux), size (vCPU + RAM), disk, region, and network configuration. Azure handles the physical host; you manage everything inside the OS.

Anatomy of an Azure VM — What Gets Created

- **VM Instance** — The actual compute resource — vCPU + RAM allocated from a physical host
- **OS Disk** — A managed disk (default) holding the operating system — persists when VM is stopped/deallocated
- **Temp Disk** — Fast local SSD on the physical host — data IS LOST when VM is stopped/deallocated
- **NIC (Network Interface Card)** — Connects the VM to a subnet in a VNet — gets a private IP
- **Public IP (optional)** — A dynamic or static IP for internet access — can be detached and reattached
- **NSG** — The firewall rules applied to the NIC or subnet

⚠ NOTE: Stopping a VM via the OS (shutdown command) does NOT deallocate it. The VM still occupies a physical host slot and you ARE billed. Use Azure Portal or CLI 'az vm deallocate' to stop billing.

VM Disk Management

Azure Managed Disks abstract the underlying storage. Three main types:

- **OS Disk** — Created automatically — Premium SSD or Standard SSD/HDD
- **Data Disk** — Additional disks you attach for application data — persists independently of the VM
- **Temp Disk** — D: drive on Windows, /dev/sdb on Linux — ephemeral, fast, but non-persistent

Disk Encryption Options

- **Server Side Encryption (SSE)** — Default — Azure encrypts data at rest on managed disks using platform-managed keys. Happens transparently with no configuration required.
- **Disk Encryption Sets** — Allow you to use customer-managed keys (CMK) stored in Azure Key Vault for SSE. You control the key rotation.
- **Azure Disk Encryption (ADE)** — Uses BitLocker (Windows) or dm-crypt (Linux) inside the VM OS. Encrypts the VM's volumes at the OS level — keys stored in Key Vault.

⚠ NOTE: SSE protects data on the physical storage. ADE protects the disk if it were detached and mounted elsewhere — adds an extra layer.

Custom Script Extensions

A VM Extension that runs a PowerShell script (Windows) or a Bash script (Linux) on a VM after deployment — automating post-deployment configuration like installing IIS, Nginx, or configuring apps.

```
# PowerShell — Install IIS via Custom Script Extension
Install-WindowsFeature -name Web-Server -IncludeManagementTools
```

`Install-WindowsFeature` → PowerShell cmdlet that installs Windows Server roles and features

`-name Web-Server` → Installs the IIS web server role

`-IncludeManagementTools` → Includes IIS Manager GUI tools alongside the core feature

⚠ NOTE: Cloud-init is the Linux equivalent for VM initialisation scripts — runs on first boot and is passed via the 'Custom Data' field during VM creation.

VM High Availability Options

2.2 Availability Sets vs Availability Zones

Both options protect VMs from failures, but at different levels of the infrastructure. Understanding the distinction is critical for the exam.

Dimension	Availability Sets	Availability Zones
Protects Against	Hardware failures in same rack / update events	Entire datacentre (zone) failure
Mechanism	Fault Domains (separate racks/power), Update Domains (staged maintenance)	Physically separate datacentres within a region
Scope	Within ONE datacentre	Across 3 datacentres in one region
Latency	Very low (same datacentre)	Low (< 2ms between zones typically)
SLA	99.95% with 2+ VMs in set	99.99% with 2+ VMs in different zones
Use Case	Older workloads, where zone support not available	New critical workloads — highest resilience
VM can be in both?	No — mutually exclusive	No — mutually exclusive

⚠ **NOTE: Fault Domains = separate physical racks with independent power and cooling. Update Domains = groups that Azure updates one at a time during planned maintenance. By default: 2 Fault Domains, 5 Update Domains.**

2.3 Azure Virtual Machine Scale Sets (VMSS)

VMSS lets you deploy and manage a set of identical, load-balanced VMs that auto-scale in and out based on metric-based rules or schedules. All VMs are created from the same base image.

Key Configuration Points

- **Orchestration Mode** — Uniform (identical VM config, required for VMSS-managed autoscale) or Flexible (mix of VM configs, recommended for new deployments)
- **Scaling Policy** — Manual, Custom (metric-based: CPU %, inbound flows), or Scheduled
- **Scale-Out** — Add VM instances when load increases — e.g., CPU > 70% for 5 mins: add 2 instances
- **Scale-In** — Remove VM instances when load decreases — e.g., CPU < 30% for 10 mins: remove 1 instance
- **Cooldown Period** — Minimum wait time between autoscale actions — prevents rapid scaling oscillation

⚠ **NOTE: Flexible Orchestration Mode is now recommended by Microsoft for new VMSS deployments. It supports a mix of VM configs and is compatible with Availability Zones.**

2.4 Azure Web Apps (App Service)

A fully managed PaaS platform for hosting web applications, REST APIs, and mobile backends. You focus on code; Azure manages the OS, runtime, patching, and scaling. Web Apps run on App Service Plans, which define the compute resources.

App Service Plan Tiers

- **Free / Shared** — Dev/test only. Shared compute. No custom domain. No SLA.
- **Basic (B1/B2/B3)** — Dedicated compute. Custom domains. Manual scale only. No autoscale.

- **Standard (S1/S2/S3)** — Autoscale (up to 10 instances). Deployment slots (up to 5). VNet integration basic.
- **Premium (P1v3/P2v3)** — Autoscale up to 30 instances. More CPU/RAM. Enhanced VNet integration.

⚠ **NOTE: Multiple Web Apps can share one App Service Plan at no extra compute cost — they share the plan's vCPUs and RAM. Be careful — noisy apps on same plan affect each other.**

Deployment Slots

Deployment Slots are live environments within a Web App (e.g., 'staging', 'production'). You deploy new code to staging, test it, then swap slots — which means staging becomes production instantly with zero downtime. If a problem is found, you can swap back immediately.

⚠ **NOTE: Swapping slots also swaps: app settings (unless marked 'Slot Setting'), connection strings, and custom domain bindings. Slot settings stay with the slot — they don't travel with the app.**

Virtual Network Integration

VNet Integration (AZ-104 topic) allows a Web App to make outbound calls to resources inside a VNet using private IPs — e.g., calling a database VM that has no public endpoint. The Web App uses a subnet in your VNet to route outbound traffic through the private network.

2.5 Azure Functions

Azure Functions is a serverless, event-driven compute service. You write individual functions — pieces of code — and Azure handles the execution environment. You are billed only when the function actually runs.

Trigger Types

- **HTTP Trigger** — Function runs when an HTTP request hits its endpoint — builds a lightweight API/webhook
- **Timer Trigger** — Runs on a CRON schedule — e.g., every night at midnight to clean up expired records
- **Blob Trigger** — Fires when a file is uploaded to Azure Blob Storage — ideal for image processing pipelines
- **Queue Trigger** — Fires when a message appears in a Storage Queue or Service Bus Queue
- **Event Grid Trigger** — Responds to Azure events published to Event Grid

⚠ **NOTE: Functions abstract away the server entirely. 'Serverless' does not mean no servers exist — it means you don't see, manage, or pay for idle server time. Azure spins compute up per invocation.**

2.6 Containers on Azure — Quick Stack

Service	What It Is	When to Use	Manages Cluster?
Docker Hub / ACR	Image registry — stores container images	Store and version container images	N/A — not a runtime
Azure Container Instances (ACI)	Run a single container directly, no orchestrator	Dev/test, short-lived jobs, simple tasks	Yes — fully managed
Azure Container Apps (ACA)	Serverless containers with auto-scale and Dapr	Microservices, event-driven, scale-to-zero	Yes — K8s hidden
Azure Kubernetes Service (AKS)	Managed K8s cluster — you manage workloads	Complex microservices needing full K8s control	Control plane managed

Azure Web App (container mode)	Deploy container image to App Service	Web app already in a container, PaaS simplicity	Yes — fully managed

⚠ **NOTE: ACI is for quick container runs. ACA is for production microservices that should scale to zero. AKS is when you need full Kubernetes flexibility and control.**

2.7 Azure Virtual Desktop

Azure Virtual Desktop (AVD) is a desktop-and-application virtualisation service running on Azure. It delivers full Windows 10/11 desktops to end users from any device using the RDP protocol over HTTPS.

- **Host Pool** — Collection of Azure VMs (session hosts) that provide the desktop sessions
- **Workspace** — Logical grouping that ties host pools and application groups together — what users see in the AVD client
- **Multi-session** — Windows 11/10 Enterprise multi-session allows multiple concurrent users on one VM — significantly reduces cost per user versus traditional VDI

⚠ **NOTE: AVD is an exam favourite for AZ-900. Know: it is a DaaS (Desktop as a Service) model. You pay per VM hour of the session host — not per user session.**

2.8 Azure Batch Service

Azure Batch runs large-scale parallel and high-performance computing (HPC) batch jobs using a managed pool of VMs. Azure Batch handles the pool provisioning, job scheduling, VM management, and retries — you provide the executable and input data.

- **Pool** — The set of compute VMs dedicated to a Batch account
- **Job** — A container for a group of tasks
- **Task** — An individual unit of work — runs a specific executable or command on one VM in the pool

⚠ **NOTE: Azure Batch Service itself has NO cost. You only pay for the underlying VMs in the pool and storage. The Batch service is essentially a job scheduler built on top of Azure VMs.**

PHASE 3 — NETWORKING SERVICES

3.1 Azure Virtual Network (VNet)

A VNet is a logically isolated network in Azure. You define the private IP address space using CIDR notation, divide it into subnets, and deploy Azure resources (VMs, App Services, etc.) into those subnets. Resources in the same VNet can communicate using private IPs by default.

VNet Fundamentals

- **Address Space** — The total CIDR block assigned to the VNet — e.g., 10.0.0.0/16 gives 65,536 addresses
- **Subnets** — Subdivisions of the VNet address space — e.g., 10.0.1.0/24 for web tier, 10.0.2.0/24 for DB tier
- **VNet is regional** — A VNet lives in one Azure region. To connect VNets in different regions: use VNet Peering or VPN Gateway
- **Private IP Allocation** — Dynamic (DHCP — can change on realloc) or Static (fixed IP assigned to NIC — stays even after deallocation)
- **Public IP** — Optional — attached to a NIC or Load Balancer. Dynamic (changes on dealloc) or Static (never changes)

⚠ NOTE: Five IP addresses in every subnet are reserved by Azure and cannot be used: .0 (network), .1 (default gateway), .2 and .3 (Azure DNS), .255 (broadcast). A /24 subnet gives you 256 - 5 = 251 usable IPs.

3.2 Network Security Groups (NSG)

NSGs are virtual firewalls applied to subnets or individual NICs. They contain a list of inbound and outbound security rules. Azure evaluates each rule in priority order — lowest number wins. There are default rules that cannot be deleted but can be overridden.

NSG Rule Properties

- **Priority** — 100–4096. Lower number = higher priority. First matching rule wins. Evaluation stops at first match.
- **Source / Destination** — IP address, CIDR range, Service Tag (e.g., 'Internet', 'VirtualNetwork'), or Application Security Group
- **Protocol** — TCP, UDP, ICMP, or Any
- **Port Range** — Single port (80), range (1024-65535), or * (all)
- **Action** — Allow or Deny

Default Rules (cannot be deleted)

```
AllowVnetInBound (65000):  Allows all traffic between resources in the same VNet
AllowAzureLoadBalancerInBound (65001):  Allows health probes from Azure Load
Balancer
DenyAllInBound (65500):  Denies all other inbound traffic — the catch-all deny
AllowVnetOutBound (65000):  Allows all outbound traffic to VNet resources
AllowInternetOutBound (65001):  Allows outbound traffic to internet
DenyAllOutBound (65500):  Denies all other outbound traffic
```

⚠ NOTE: NSG on a subnet applies to ALL NICs in that subnet. NSG on a NIC applies only to that specific VM. When both subnet-NSG and NIC-NSG exist, traffic must pass BOTH for inbound, and pass NIC-NSG first then subnet-NSG for outbound.

3.3 Application Security Groups (ASG)

ASGs allow you to group VMs logically (e.g., 'WebServers', 'DatabaseServers') and use those group names in NSG rules instead of IP addresses. When you add a VM to an ASG, its NIC is a member. When the VM's IP changes, no NSG rule update is needed.

⚠ NOTE: ASGs work only within the same VNet. The key benefit: you manage security rules based on workload identity (group name), not IP addresses that change frequently.

3.4 Virtual Network Peering

VNet Peering connects two VNets — resources in both VNets can communicate using private IPs as if they were in the same VNet. Traffic flows through Azure's backbone network, not the internet.

- **Regional Peering** — Connects VNets in the same Azure region
- **Global Peering** — Connects VNets across different Azure regions
- **Non-Transitive** — If VNet A is peered with VNet B, and VNet B is peered with VNet C, VNet A CANNOT talk to VNet C via the peering. Transitive routing must be configured explicitly via a hub VNet with gateway transit.

⚠ NOTE: Peering is NOT free. You are billed per GB of data transferred across the peer. Peering must be configured in BOTH directions — A→B and B→A are separate peering connections.

3.5 Azure Bastion

Azure Bastion is a PaaS service deployed inside your VNet (in a subnet named exactly 'AzureBastionSubnet') that provides browser-based RDP and SSH access to VMs via the Azure Portal over TLS port 443. No public IP is needed on the target VM.

- **Basic SKU** — HTML5 browser-based RDP/SSH. Fixed max concurrent sessions.
- **Standard SKU** — Includes native client support (connect via local RDP/SSH client, not just browser), shareable links, host scaling.

⚠ NOTE: AzureBastionSubnet must be at least a /26 or larger. The subnet must be named exactly 'AzureBastionSubnet' — any other name will be rejected by Azure.

3.6 Azure Load Balancer

Azure Load Balancer operates at Layer 4 (TCP/UDP) and distributes inbound traffic across a pool of backend VMs using configured load balancing rules and health probes.

Feature	Basic SKU	Standard SKU
Availability Zone support	No	Yes — zone-redundant frontend
VMSS backend support	Limited	Full support
Outbound SNAT	Yes	Yes — explicit outbound rules required
Health probe protocols	TCP, HTTP	TCP, HTTP, HTTPS
SLA	No SLA	99.99%
Cost	Free	Billed per rule/hour + data processed
Recommendation	Dev/test only	All production workloads

⚠ NOTE: Inbound NAT Rules let you map a public port on the Load Balancer frontend to a specific port on a specific backend VM. E.g., public:50001 → VM1:3389 for direct RDP to VM1 through the LB.

3.7 User Defined Routes (UDR) [AZ-104]

By default, Azure uses system routes to handle traffic between subnets, to the internet, and to on-premises. User Defined Routes allow you to override these system routes. The most common use case is routing all outbound internet traffic through an Azure Firewall or NVA (Network Virtual Appliance) for inspection.

```
Route Table:  Resource containing custom routes — associated with a subnet
Next Hop Types:  VirtualNetworkGateway | VirtualAppliance (for Firewall/NVA) |
VnetLocal | Internet | None
```

⚠ **NOTE: 'None' as next hop drops packets — effectively a blackhole route. This is used to block specific traffic paths. A UDR on the GatewaySubnet pointing to an NVA controls VPN traffic.**

3.8 Network Watcher [AZ-104]

Network Watcher is a suite of network diagnostic tools for Azure VNets.

Tool	What It Does	Best For
Connection Troubleshoot	Tests TCP connectivity between source and destination VMs	Debugging VM-to-VM connectivity failures
Connection Monitor	Continuous monitoring of network paths with latency metrics	Ongoing health monitoring of critical paths
IP Flow Verify	Checks if a specific packet (IP, port, protocol, direction) is allowed or denied	Quickly finding the NSG rule blocking traffic
Next Hop	Shows the effective next hop for traffic from a VM to a destination IP	Debugging routing issues — is traffic going where expected?
NSG Diagnostic	Shows all effective NSG rules applied to a VM or NIC	Understanding the combined effect of multiple NSGs
NSG Flow Logs	Logs all allowed/denied traffic flows through an NSG to Storage Account	Traffic auditing and forensics

3.9 DNS Services in Azure

Azure Public DNS

Host your public domain's DNS records in Azure DNS for fast, reliable name resolution. Azure acts as the authoritative DNS for your domain. You point your domain's NS records at Azure DNS name servers.

Azure Private DNS

A managed DNS zone for private name resolution within VNets. Resources in linked VNets can resolve names in the private zone without a DNS server. Supports auto-registration — VMs automatically register their hostname.

⚠ **NOTE: Private DNS Zones are global resources — not regional. You link them to VNets. One VNet can be linked to many zones. One zone can be linked to many VNets.**

3.10 Traffic Distribution Services — Comparison [AZ-900]

Service	Layer	Scope	Use Case	Key Feature

Azure Load Balancer	L4 TCP/UDP	Regional	Distribute traffic across VMs in same region	Health probes + SNAT
Azure Traffic Manager	DNS (L7)	Global	Route users to best regional endpoint	Routing policies: Performance, Weighted, Priority, Geographic
Azure Front Door	L7 HTTP	Global	CDN + global web app delivery + WAF	Edge caching + SSL offload + WAF
Azure Application Gateway	L7 HTTP	Regional	Advanced web traffic with URL-based routing + WAF	Path-based routing, cookie affinity, SSL termination

⚠ **NOTE: Application Gateway was in the 104 course infrastructure but not explicitly listed as a lecture. Know it conceptually: it's a regional Layer 7 LB with WAF. Front Door is the global equivalent.**

3.11 VPN Gateway vs ExpressRoute [AZ-900]

Aspect	VPN Gateway	ExpressRoute
Connection Type	Encrypted tunnel over public internet (IPSec)	Private dedicated connection via telco provider
Bandwidth	Up to ~10 Gbps (VpnGw5)	Up to 100 Gbps
Reliability	Internet-dependent — variable latency	Dedicated circuit — consistent low latency, SLA-backed
Setup Time	Hours (self-serve in portal)	Weeks to months (requires provider provisioning)
Cost	Lower — no provider circuit fee	Higher — circuit + port fees
Use Case	Branch office connectivity, dev/test, remote access	Enterprise, financial, high-bandwidth, compliance-driven
Data over internet?	Yes (encrypted)	Never — private MPLS path

PHASE 4 — STORAGE SERVICES

4.1 Azure Storage Account — Foundation

A Storage Account is the top-level container that groups all Azure storage services. It provides a unique namespace (accountname.blob.core.windows.net etc.) and defines the billing scope, redundancy type, and performance tier.

Storage Account Types

- **General Purpose v2 (GPv2)** — Default and recommended. Supports Blob, File, Queue, and Table storage. All redundancy options. Supports access tiers.
- **Blob Storage** — Legacy type. Only supports Blob. Migrate to GPv2.
- **Premium Block Blob** — High-performance SSDs for block blobs. Low latency for high-transaction workloads. LRS/ZRS only.
- **Premium File Shares** — High-performance SSDs for Azure Files. Billed per provisioned capacity (not used).
- **Premium Page Blob** — For page blobs — used as VHDs for VM disks. Ultra-low latency.

⚠ **NOTE: Premium Storage Accounts CANNOT use access tiers (Hot/Cool/Archive). They are always 'always-on' SSDs — you pay a premium for the performance, not for access frequency.**

4.2 Data Redundancy Options

Redundancy defines how many copies of your data Azure keeps and where. More redundancy = higher durability + higher cost.

Redundancy	Copies	Where	Survives	Use Case
LRS — Locally Redundant	3	Same datacentre (3 racks)	Rack/drive failure	Dev/test, cheap non-critical data
ZRS — Zone Redundant	3	3 Availability Zones (same region)	Single zone (datacentre) failure	High-availability production — recommended
GRS — Geo Redundant	6	3 LRS primary + 3 LRS secondary region	Primary region failure	Business continuity — critical data
GZRS — Geo-Zone Redundant	6	3 ZRS primary + 3 LRS secondary region	Zone + primary region failure	Maximum redundancy
RA-GRS	6 + readable secondary	Same as GRS	Primary region failure	GRS + read access to secondary region

⚠ **NOTE: By default, data in secondary region (GRS/GZRS) is NOT readable. With RA-GRS, you get a secondary read endpoint (.secondary.blob.core.windows.net) for geo-redundant reads.**

4.3 Azure Blob Storage — Deep Dive

Blob (Binary Large Object) storage is the most commonly used Azure Storage service. It stores unstructured data — any binary file, document, image, video, log file, backup. Data is stored in containers (analogous to folders) as named blobs.

Blob Types

- **Block Blob** — Default type. For storing text and binary files. Made of blocks that can be uploaded and committed independently. Up to ~190 TB.

- **Append Blob** — Optimised for append operations — new data added to the end only. Ideal for logging and audit trails.
- **Page Blob** — Optimised for random read/write operations. Used internally for Azure VM disks (VHDs). 512-byte pages. Up to 8 TB.

Access Tiers

Tier	Storage Cost	Access Cost	Min Storage Duration	Best For
Hot	Highest	Lowest	None	Frequently accessed data — active app assets
Cool	Lower	Higher	30 days	Infrequent access — backups, older datasets
Cold	Lower than Cool	Higher than Cool	90 days	Rarely accessed data archived for compliance
Archive	Lowest (~10% of Hot)	Very high + rehydration delay	180 days	Long-term archival — rarely retrieved

⚠ **NOTE: Archive tier blobs CANNOT be read directly. They must first be 'rehydrated' to Hot or Cool — which takes up to 15 hours on standard priority. This is a common exam question.**

Blob Lifecycle Management

Lifecycle Management policies automatically transition blobs between access tiers or delete them based on age rules — eliminating the need to manually manage tier migrations.

```
# Example Lifecycle Policy Rule (JSON)
{
  "rules": [{
    "name": "archiveRule",
    "type": "Lifecycle",
    "definition": {
      "filters": { "blobTypes": ["blockBlob"] },
      "actions": {
        "baseBlob": {
          "tierToCool": { "daysAfterModificationGreaterThan": 30 },
          "tierToArchive": { "daysAfterModificationGreaterThan": 90 },
          "delete": { "daysAfterModificationGreaterThan": 365 }
        }
      }
    }
  }]
}
```

Blob Authorization Methods

Method	What It Is	Best For	Expiry?
Storage Access Keys	2 root keys — full access to everything in account	Internal server-side access only — NEVER in client apps	No — permanent until rotated
Shared Access Signature (SAS)	Time-limited token granting scoped permissions (read blob, write container, etc.)	Granting temporary access to specific resources	Yes — mandatory expiry

Stored Access Policy	Server-side policy defining SAS parameters — SAS tokens reference it	Revocable SAS — update policy to invalidate all SAS tokens	Policy-defined
Azure AD / RBAC	Entra ID identity-based access using Storage Blob Data Reader/Contributor roles	Recommended for internal Azure resource access	Based on role assignment

⚠ **NOTE: SAS token = signed URL. If the token is leaked, anyone with the URL gets that access until expiry. Use Stored Access Policy to be able to revoke SAS tokens before they expire.**

Immutable Blob Storage

Immutable storage enforces Write Once, Read Many (WORM) on blobs. Used for compliance scenarios (financial records, legal documents). Two policy types:

- **Time-Based Retention Policy** — Blobs cannot be modified or deleted for a specified duration after creation
- **Legal Hold Policy** — Blobs are locked indefinitely until the hold is explicitly cleared — used during legal investigations

4.4 Azure Files — File Shares

Azure Files provides fully managed SMB (Server Message Block) and NFS file shares accessible from Windows, Linux, and macOS. Mount them as network drives — exactly like an on-premises NAS.

Key Features

- **SMB 3.0** — Windows native protocol — mount as Z: drive
- **NFS 4.1** — Linux/macOS native — mount as /mnt/share
- **File Share Snapshots** — Point-in-time read-only copy of a file share. Manual or scheduled. Used for accidental deletion recovery.
- **Soft Delete** — Deleted files and shares kept for a configurable retention period (1-365 days) — recoverable after accidental deletion
- **Azure File Sync** — Synchronise on-premises Windows Server with Azure File Share. Cloud tiering moves infrequently accessed files to Azure automatically.

⚠ **NOTE: Azure Files requires port 445 (SMB) to be open for on-premises mounting. Many ISPs block this port. For blocked port 445, use Azure File Sync agent or access files via Azure portal/Storage Explorer only.**

4.5 Storage Tools

- **Azure Storage Explorer** — Free GUI desktop application (Windows/macOS/Linux) for browsing and managing Storage Account contents — blobs, files, queues, tables
- **AzCopy** — Command-line tool for high-speed, resumable transfers to/from Storage Accounts. Supports batch operations, synchronisation, and service-to-service copy.
- **Azure Data Box** — Physical appliance ordered via portal for offline bulk data migration. Microsoft ships the device, you copy data locally, ship it back, Microsoft uploads to Azure. For 40TB+ transfers where internet is impractical.

4.6 Storage Service Endpoints & Private Endpoints [AZ-104]

Service Endpoints

Service Endpoints extend VNet identity to Azure services — traffic from a subnet to an Azure Storage Account flows over the Azure backbone network instead of the public internet. The storage account's firewall can then restrict access to only that subnet.

Private Endpoints

A Private Endpoint assigns a private IP address from your VNet directly to an Azure service (e.g., Storage Account). Requests resolve to this private IP — data never touches the internet at all. This is stronger than Service Endpoints.

Aspect	Service Endpoint	Private Endpoint
Traffic path	Azure backbone (still exits VNet)	Stays within VNet via private IP
Accessible from internet?	Can still allow public access	Fully blocks public access when configured
DNS resolution	Still public FQDN	Private DNS zone maps FQDN → private IP
Cost	Free	Billed per hour + per GB processed
Recommended for?	Simple VNet restriction	High-security, compliance-driven scenarios

PHASE 5 — DATABASE SERVICES [AZ-900]

5.1 Azure SQL Database

A fully managed relational database based on Microsoft SQL Server. Azure handles HA, patching, backups, and scaling. You get a SQL Server-compatible endpoint and can start querying with T-SQL immediately.

Purchasing Model	DTU Model	vCore Model
What is it?	Bundled CPU+IO+Memory units	Choose vCPU count and memory separately
Predictable?	Yes — fixed resource bundle	More granular — choose resources independently
Serverless?	No	Yes — Serverless tier scales compute to zero when idle
Best For	Simpler workloads, easy comparison	Production workloads, Azure Hybrid Benefit (SQL Server licence)

5.2 Azure Cosmos DB

Cosmos DB is a globally distributed, multi-model NoSQL database with guaranteed single-digit millisecond read latency. It supports multiple APIs: SQL (Core), MongoDB, Cassandra, Gremlin (Graph), and Table.

Key Concepts

- **Throughput** — Measured in RU/s (Request Units per second) — 1 RU = cost to read a 1 KB item. Provisioned or serverless.
- **Partition Key** — Determines how data is distributed across physical partitions. Choosing a good partition key is critical for even distribution and performance.
- **Consistency Levels** — Five levels from Strong (highest consistency, lowest availability) to Eventual (highest availability, potential stale reads). Azure-unique: Session consistency is default.
- **Global Distribution** — Add or remove Azure regions with one click — data is replicated automatically. Reads from nearest region.

⚠ **NOTE: Cosmos DB is NOT just for documents. It is a multi-model engine. The SQL API does NOT mean it runs SQL Server — it means you use SQL-like query syntax against JSON documents.**

5.3 Data Engineering Stack [AZ-900 Overview]

Service	Role in Pipeline	Key Technology
Azure Data Lake Storage Gen2	Raw data ingestion and storage (Bronze layer)	Blob Storage with hierarchical namespace
Azure Data Factory	ETL/ELT pipeline orchestration	No-code + code pipelines
Azure Synapse Analytics	Data warehouse + analytical queries	MPP SQL engine + Apache Spark
Azure Databricks	Big data transformation + ML	Apache Spark (created by Databricks founders)
Microsoft Fabric	Unified data platform (all-in-one)	Lakehouse + Warehouse + Power BI integrated
Power BI	Business Intelligence and reporting	DAX + M query language

PHASE 6 — IDENTITY AND ACCESS MANAGEMENT

6.1 Microsoft Entra ID (formerly Azure Active Directory)

Microsoft Entra ID is Azure's cloud-based identity and access management service. It is the directory of users, groups, service principals, and applications. Every authentication event in Azure — portal login, CLI command, API call — goes through Entra ID.

⚠ **NOTE: Microsoft rebranded Azure Active Directory (Azure AD) to Microsoft Entra ID in 2023. Both names appear in exams. They refer to the same service.**

Core Concepts

- **Tenant** — A dedicated instance of Entra ID for one organisation — created automatically when you sign up for Azure or Microsoft 365
- **Users** — Individual identities (employees, guests). Can be cloud-only or synced from on-premises AD via Entra Connect.
- **Groups** — Collections of users for bulk role/licence assignment. Security groups (for RBAC) vs Microsoft 365 groups (for collaboration).
- **Service Principal** — The identity of an application — like a 'machine user' for apps to authenticate to Azure services
- **Managed Identity** — An Azure resource's auto-managed service principal — no credentials needed; Azure handles it. System-assigned (tied to resource lifecycle) or user-assigned (standalone).

Entra ID Licence Tiers

Feature	Free	P1	P2
Basic SSO, MFA	Yes	Yes	Yes
Conditional Access	No	Yes	Yes
Self-Service Password Reset	No	Yes	Yes
Group-Based Licensing	No	Yes	Yes
Privileged Identity Management (PIM)	No	No	Yes
Identity Protection (risk-based CA)	No	No	Yes

6.2 Role-Based Access Control (RBAC)

RBAC is the authorisation system for Azure resources. You assign a role (set of permissions) to an identity (user, group, or service principal) at a specific scope.

The Three Parts of an RBAC Assignment

```
Security Principal:  WHO: User, Group, Service Principal, or Managed Identity
Role Definition:  WHAT: A collection of permissions. Built-in or custom.
Scope:  WHERE: Management Group > Subscription > Resource Group > Resource
```

Key Built-in Roles

Role	Can Read?	Can Create/Modify?	Can Delete?	Can Grant Access?
Owner	Yes	Yes	Yes	Yes — full control

Contributor	Yes	Yes	Yes	No — cannot modify access
Reader	Yes	No	No	No
User Access Administrator	Yes	No (resources)	No	Yes — only manages access, not resources

⚠ NOTE: RBAC is additive — if a user is in two groups with different roles on the same resource, both roles apply (union of permissions). DENY assignments (at Azure Policy level) override ALLOW.

Custom Roles [AZ-104]

When built-in roles are too broad or too narrow, you create Custom Roles with exactly the actions you want. A custom role defines: Actions (allowed operations), NotActions (excluded from Actions), DataActions (data plane ops like read blob), and AssignableScopes.

```
# Custom Role JSON structure (abbreviated)
{
  "Name": "VM Operator — Restart Only",
  "Actions": [
    "Microsoft.Compute/virtualMachines/read",
    "Microsoft.Compute/virtualMachines/restart/action"
  ],
  "NotActions": [],
  "AssignableScopes": ["/subscriptions/{subscriptionId}"]
}
```

⚠ NOTE: NotActions are NOT explicit Deny. They subtract from Actions. If another role assignment grants the action, the user can still do it. Use Azure Policy for true Deny.

6.3 Self-Service Password Reset (SSPR) [AZ-104]

SSPR lets non-admin users reset their own Entra ID passwords without contacting IT helpdesk. Requires: Entra ID P1 or P2 licence, at least one registered authentication method (email, phone, authenticator app).

⚠ NOTE: SSPR does NOT work for admin accounts by default. Admin accounts have a stricter policy requiring more authentication methods. SSPR only applies to cloud-synced passwords, not on-premises AD passwords, unless write-back is enabled.

6.4 Conditional Access Policies [AZ-900 P1 Feature]

Conditional Access is the 'if-then' policy engine of Entra ID. It evaluates signal data at sign-in and enforces controls. Signals: user identity, location, device compliance, app being accessed, risk score. Controls: require MFA, block access, require compliant device.

- **Example Policy** — IF user logs in from outside the company network AND is accessing the Azure portal → THEN require MFA
- **Named Locations** — Define trusted IP ranges — use in CA policies to distinguish corporate network from elsewhere

⚠ NOTE: Conditional Access requires Entra ID P1 licence minimum. Without P1, MFA can only be enforced globally (for all logins), not conditionally.

6.5 Microsoft Entra Domain Services [AZ-900]

Entra Domain Services provides a managed set of traditional Active Directory features — domain join, Group Policy, LDAP, Kerberos/NTLM authentication — without running or patching Domain Controller VMs. Azure provisions two managed DCs (invisible to you) and syncs identities from Entra ID.

⚠ **NOTE: Entra Domain Services is a ONE-WAY sync from Entra ID DOWN. Changes in Entra Domain Services do not sync back to Entra ID or on-premises AD. It is a managed service — you cannot directly access the DC VMs.**

PHASE 7 — INFRASTRUCTURE AS CODE (ARM, Bicep, CLI, PowerShell)

7.1 Azure Resource Manager (ARM) — The Control Plane

Every operation on Azure — whether through the Portal, CLI, PowerShell, or REST API — ultimately goes through Azure Resource Manager. ARM is the management layer that authenticates requests, authorises them against RBAC, and processes resource operations. ARM Templates are declarative JSON files that define the desired state of Azure resources.

ARM Template Structure

```
{
  "$schema": "https://schema.management.azure.com/schemas/...",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "storageAccountName": {
      "type": "string",
      "metadata": { "description": "Storage account name" }
    }
  },
  "variables": {
    "storageAccountId": "[resourceId('Microsoft.Storage/storageAccounts',
parameters('storageAccountName'))]"
  },
  "resources": [
    {
      "type": "Microsoft.Storage/storageAccounts",
      "apiVersion": "2021-09-01",
      "name": "[parameters('storageAccountName')]",
      "location": "[resourceGroup().location]",
      "sku": { "name": "Standard_LRS" },
      "kind": "StorageV2"
    }
  ],
  "outputs": {
    "storageEndpoint": {
      "type": "string",
      "value": "[reference(variables('storageAccountId')).primaryEndpoints.blob]"
    }
  }
}
```

`$schema` → Points to the ARM template schema definition — tells VS Code what is valid

`contentVersion` → Version of your template for tracking purposes — always '1.0.0.0'

`parameters` → Values supplied at deployment time — makes template reusable across environments

`variables` → Computed values internal to the template — reduces repetition

`resources` → The actual Azure resources to create/update — the core of the template

`outputs` → Values emitted after deployment — e.g., endpoint URLs or resource IDs for use in pipelines

⚠ NOTE: ARM Templates are idempotent — deploying the same template twice produces the same result. They use Incremental mode (add/update only) or Complete mode (deletes resources NOT in template) by default Incremental.

7.2 Bicep — Simplified ARM

Bicep is a domain-specific language (DSL) that compiles to ARM JSON. It is the recommended modern approach for writing Azure IaC. It offers concise syntax, type safety, native module support, and deep VS Code integration.

Same Storage Account in Bicep

```
param storageAccountName string
```

```
resource storageAccount 'Microsoft.Storage/storageAccounts@2021-09-01' = {
  name: storageAccountName
  location: resourceGroup().location
  sku: { name: 'Standard_LRS' }
  kind: 'StorageV2'
}

output storageEndpoint string = storageAccount.properties.primaryEndpoints.blob
```

`param storageAccountName string` → Declares a parameter of type string — passed at deployment time

`resource storageAccount '...' = {` → Declares a resource with symbolic name 'storageAccount' — used for referencing elsewhere in file

`'Microsoft.Storage/storageAccounts@2021-09-01'` → Resource type @ API version — tells ARM which schema to use

`location: resourceGroup().location` → Built-in function returns the resource group's region — no hardcoding needed

`output storageEndpoint` → Emits the blob endpoint — references the deployed resource's live properties

Bicep Modules [AZ-104]

Bicep modules allow you to split your infrastructure into reusable, composable files. A module is any .bicep file called from a main.bicep file. This enables building a library of reusable components (network module, VM module, storage module).

```
// main.bicep — calling a module
module vnetModule './modules/vnet.bicep' = {
  name: 'vnetDeployment'
  params: {
    vnetName: 'prod-vnet'
    addressPrefix: '10.0.0.0/16'
  }
}

// Use output from module in parent
output vnetId string = vnetModule.outputs.vnetId
```

`module vnetModule './modules/vnet.bicep' = {` → Declares a module reference — './modules/vnet.bicep' is the path to the reusable module file

`name: 'vnetDeployment'` → Deployment name for this module instance — appears in Azure deployment history

`params: { ... }` → Values passed into the module's declared parameters — modules have their own param blocks

`vnetModule.outputs.vnetId` → Access an output from the module — modules must declare outputs to expose values to the parent

7.3 Azure CLI — Key Commands

```
# Login and set context
az login
az account set --subscription 'My Subscription'

# Resource Group
az group create --name myRG --location eastus

# Virtual Network
az network vnet create --name myVNet --resource-group myRG --address-prefix 10.0.0.0/16
az network vnet subnet create --name mySubnet --vnet-name myVNet --resource-group myRG --
address-prefix 10.0.1.0/24

# Network Security Group
az network nsg create --name myNSG --resource-group myRG
az network nsg rule create --name AllowHTTP --nsg-name myNSG --resource-group myRG --
priority 100 --access Allow --protocol TCP --destination-port-ranges 80

# Public IP
```

```
az network public-ip create --name myPublicIP --resource-group myRG --allocation-method
Static

# Network Interface
az network nic create --name myNIC --resource-group myRG --vnet-name myVNet --subnet
mySubnet

# Virtual Machine
az vm create --name myVM --resource-group myRG --image Win2022Datacenter --admin-username
azureuser --nics myNIC

# Storage Account
az storage account create --name mystorageacct --resource-group myRG --location eastus --
sku Standard_LRS

# Deploy ARM Template
az deployment group create --resource-group myRG --template-file template.json --
parameters @params.json
```

PHASE 8 — GOVERNANCE AND MANAGEMENT

8.1 Azure Resource Hierarchy

Azure resources are organised in a strict hierarchy. Policies and RBAC assigned at a higher level are inherited downward.

```
Management Groups:  Top level — groups subscriptions. Apply governance across
multiple subscriptions simultaneously.
Subscriptions:  Billing + access boundary. Resources are created within
subscriptions. Separate prod/dev/test by subscription.
Resource Groups:  Logical container for related resources. Resources in same
group share lifecycle — delete group = delete all resources.
Resources:  Individual services — VM, VNet, Storage Account, etc.
```

⚠ NOTE: A resource can only belong to ONE resource group at a time. Moving a resource to another resource group is supported but causes a brief unavailability in some services.

8.2 Azure Policy

Azure Policy evaluates resources against defined rules (policy definitions) and enforces compliance. It can deny non-compliant resources at creation, audit existing non-compliant resources, or auto-remediate them.

Effect Types

- **Deny** — Blocks the create/update operation if resource doesn't comply — most restrictive
- **Audit** — Allows creation but marks resource as non-compliant in the compliance dashboard — good for starting governance
- **AuditIfNotExists** — Audit a related resource if it doesn't exist — e.g., flag VM if it has no Backup enabled
- **DeployIfNotExists** — Automatically deploy a related resource if it doesn't exist — e.g., deploy Log Analytics agent on new VMs
- **Modify** — Add/modify tags or properties on resources at creation or remediation

⚠ NOTE: Azure Policy Initiative = a set of policies grouped together. E.g., the 'Azure Security Benchmark' initiative contains 200+ policies. Assigning one initiative is better than assigning 200 individual policies.

8.3 Resource Locks [AZ-104]

Resource Locks prevent accidental modification or deletion of resources. They override any RBAC permissions — even an Owner cannot delete a locked resource without first removing the lock.

Lock Type	Prevents Delete?	Prevents Modify?	Who Can Remove Lock?
CanNotDelete (Delete Lock)	Yes	No — modifications allowed	Owner or User Access Administrator
ReadOnly (Read-Only Lock)	Yes	Yes — no modifications allowed	Owner or User Access Administrator

⚠ NOTE: ReadOnly lock on a resource group means you cannot create or delete resources within it, and cannot modify existing ones. This can break some operations like creating a VM (which modifies the RG).

8.4 Resource Tags

Tags are name/value pairs applied to Azure resources and resource groups for organising, billing allocation, and policy targeting. A resource can have up to 50 tags. Tags are NOT inherited from resource group to resources automatically.

- **Billing Allocation** — Tag resources with cost centre or project codes — filter Azure Cost Management by tag
- **Environment Classification** — Environment=Production, Environment=Development — target Azure Policy to specific environments
- **Automation** — Tag VMs with Shutdown=18:00 — an Azure Automation runbook reads tags to shut down after-hours

⚠ **NOTE: Tags on a resource group do NOT automatically apply to resources in that group. Use Azure Policy 'Inherit a tag from the resource group' to enforce tag inheritance.**

8.5 Management Groups [Both]

Management Groups sit above subscriptions in the hierarchy. You can apply RBAC and Azure Policy at the Management Group level — which is then inherited by all subscriptions, resource groups, and resources beneath it.

- **Root Management Group** — Every Entra ID tenant has one root management group. Applying policy here affects ALL subscriptions in the tenant.
- **Nesting** — Management groups can be nested up to 6 levels deep below the root
- **Use Case** — Apply 'Allowed Locations = UK South, UK West' policy at MG level to enforce data residency for all UK business unit subscriptions

PHASE 9 — MONITORING AND BACKUP/DISASTER RECOVERY

9.1 Azure Monitor — Overview

Azure Monitor is the overarching platform for collecting, analysing, and acting on telemetry from Azure and on-premises resources. It collects two data types: Metrics (numerical time-series values, e.g., CPU%) and Logs (structured/unstructured events stored in Log Analytics Workspace).

Azure Monitor Components

- **Metrics** — Lightweight near-real-time numerical data. Stored for 93 days. Viewable in Metrics Explorer. Used for dashboards and alerts.
- **Logs (Log Analytics)** — Rich query-capable log data. Stored in Log Analytics Workspace. Queried with KQL. Used for deep analysis and alerts.
- **Alerts** — Rules that fire when a condition is met — metric crosses a threshold or a log query returns results.
- **Action Groups** — Define what happens when an alert fires — email, SMS, push notification, webhook, Azure Function, Logic App, Runbook.
- **VM Insights** — Pre-built dashboards for VM performance (CPU, memory, disk, network) and dependency mapping — auto-deploys the monitoring agent.

9.2 Log Analytics Workspace

Log Analytics Workspace is the backend store and query engine for Azure Monitor logs. All logs from VMs, Azure services, and security tools land here. You use KQL (Kusto Query Language) to query them.

KQL Query Examples

```
// Find all VMs with CPU > 80% in last 24 hours
Perf
| where TimeGenerated > ago(24h)
| where CounterName == '% Processor Time'
| where CounterValue > 80
| summarize avg(CounterValue) by Computer, bin(TimeGenerated, 1h)
| order by avg_CounterValue desc

// Find all failed login attempts
SecurityEvent
| where EventID == 4625
| where TimeGenerated > ago(7d)
| summarize count() by Account, IpAddress
| order by count_ desc
```

`| where` → Filters rows — like SQL WHERE clause

`| summarize` → Aggregates data — like SQL GROUP BY

`| order by` → Sorts results — like SQL ORDER BY

`ago(24h)` → Built-in time function — returns datetime 24 hours ago

`bin(TimeGenerated, 1h)` → Groups timestamps into 1-hour buckets for time-series aggregation

9.3 Azure Backup [AZ-104]

Azure Backup is a built-in Azure service for backing up and restoring data. All backups are managed through a Recovery Services Vault — a storage entity that holds backup data and defines backup policies.

What Can Be Backed Up?

- **Azure Virtual Machines** — Full VM backup via snapshot — stored in vault. Includes all disks.
- **Azure File Shares** — Backup entire file share contents to vault

- **Azure SQL in VMs** — Workload-aware SQL backup — transaction log backups for low RPO
- **On-premises Files & Folders** — MARS (Microsoft Azure Recovery Services) agent installed on Windows Server — backs up selected files to vault
- **On-premises VMs** — Azure Backup Server or System Center DPM for full on-premises VM backup to vault

Recovery Types for VMs

- **File Restore** — Mount the backup snapshot as a drive — restore individual files without restoring the whole VM
- **VM Restore** — Restore the entire VM from a backup point — creates a new VM or overwrites the existing one
- **Disk Restore** — Restore individual disks only — attach to an existing VM

Azure Backup Vault vs Recovery Services Vault

Aspect	Recovery Services Vault	Backup Vault
Supports Azure VMs?	Yes — full VM + disk backup	Yes — managed disk backup
Supports Azure Files?	Yes	No
MARS Agent (on-prem)?	Yes	No
Azure Blobs?	No	Yes — operational blob backup
Azure PostgreSQL?	No	Yes
Primary Use Case	VMs, Files, on-prem backup — most workloads	Simplified managed disk + blob backup

⚠ **NOTE: If in doubt — use Recovery Services Vault. It is the primary, feature-rich option. Backup Vault is a simplified option for specific newer services.**

9.4 Azure Site Recovery (ASR) [AZ-104]

ASR is Azure's Disaster Recovery as a Service. It continuously replicates VM disk changes (using change-block tracking) from a source region to a target region. In a DR event, you fail over — the replicated VM boots in the target region.

Key Metrics

- **RPO — Recovery Point Objective** — How much data could be lost (time since last replication sync). ASR: typically < 1 minute.
- **RTO — Recovery Time Objective** — How long to restore service. ASR: VMs can be started in target region within minutes of failover trigger.

⚠ **NOTE: ASR replicates CONTINUOUSLY. Azure Backup takes periodic snapshots (daily). For critical VMs where you cannot lose hours of data: use ASR. For DR with relaxed RPO: Backup is sufficient.**

Failover Types

- **Test Failover** — Boots VM in target region in an isolated VNet — no impact to production. Used for DR drills.
- **Planned Failover** — Manual, controlled failover — used for planned migration or maintenance. Source VM shut down first to ensure zero data loss.
- **Unplanned Failover** — Emergency failover during an actual outage — source region unavailable. VM booted with last available replication point.

PHASE 10 — SECURITY SERVICES [AZ-900]

10.1 Microsoft Defender for Cloud

Defender for Cloud is a unified Cloud Security Posture Management (CSPM) and Cloud Workload Protection Platform (CWPP). Free tier provides basic CSPM. Enhanced plans add real-time threat protection per workload type.

- **Secure Score** — A percentage (0-100%) representing your security posture. Each recommendation, when implemented, increases the score. Drive decisions by score improvement impact.
- **Security Recommendations** — Prioritised list of improvements — e.g., 'Enable MFA for accounts with Owner permissions', 'Restrict SSH access on VMs'
- **Workload Protection** — Defender for Servers, Defender for SQL, Defender for Storage, Defender for Containers — real-time threat detection per service type
- **Multi-Cloud** — Defender for Cloud also supports AWS and GCP workloads — centralised security posture across clouds

10.2 Microsoft Sentinel

Sentinel is Azure's cloud-native SIEM (Security Information and Event Management) and SOAR (Security Orchestration, Automation and Response) platform. It collects data from connectors, runs analytics rules to detect threats, creates incidents, and can run automated playbooks to respond.

Feature	Description
Data Connectors	Pre-built integrations: Azure AD, Microsoft 365, Defender products, Palo Alto, AWS CloudTrail, etc.
Analytics Rules	KQL-based detection rules that generate alerts/incidents when suspicious patterns are found
Incidents	Correlated group of alerts representing a potential attack — investigated by security analysts
Workbooks	Interactive visual dashboards built on Log Analytics data — pre-built and custom
Playbooks	Azure Logic Apps triggered by incidents — automate response (block IP, disable user, send Teams alert)
UEBA	User and Entity Behaviour Analytics — baseline normal behaviour, detect anomalies

10.3 Azure Key Vault

Key Vault provides secure storage and access control for three types of assets: Secrets (passwords, connection strings), Keys (RSA/EC cryptographic keys for encryption), and Certificates (SSL/TLS certificates with auto-renewal).

⚠ NOTE: The primary use pattern in modern Azure architecture: apps use Managed Identity to authenticate to Key Vault and fetch secrets at runtime. No secrets in code, config files, or environment variables. All access logged to Azure Monitor.

10.4 Zero Trust Model [AZ-900]

Zero Trust is a security model based on the principle: 'Never trust, always verify.' Instead of assuming anything inside the corporate network is safe, Zero Trust requires verification of every request regardless of origin.

- **Verify Explicitly** — Always authenticate and authorise based on all available signals: identity, location, device health, service/workload, data classification, anomalies
- **Use Least Privilege Access** — Limit user access with Just-In-Time and Just-Enough-Access (JIT/JEA), risk-based adaptive policies, and data protection
- **Assume Breach** — Minimise blast radius, end-to-end encryption, use analytics to get visibility and drive threat detection

⚠ **NOTE: In Azure, Zero Trust is implemented through Conditional Access (Entra ID), RBAC (least privilege), Defender for Cloud (assume breach), Private Endpoints (no public network trust), and Azure Firewall/NSGs.**

PHASE 11 — OTHER AZURE SERVICES [AZ-900]

11.1 Azure DevOps

Azure DevOps is a suite of tools for implementing DevOps practices — planning, source control, CI/CD, and testing — all integrated with Azure deployment targets.

Tool	What It Does	Equivalent
Azure Boards	Agile project management — epics, user stories, tasks, sprints, Kanban	Jira
Azure Repos	Git-based source code hosting — private repos	GitHub / GitLab
Azure Pipelines (CI)	Build pipeline — triggered on commit; compiles, tests, creates artifact	Jenkins / GitHub Actions
Azure Pipelines (CD)	Release pipeline — deploys artifact to Azure Web App, AKS, VM, etc.	Jenkins / GitHub Actions deploy stage
Azure Test Plans	Manual and exploratory testing management	Zephyr / qTest
Azure Artifacts	Package management — npm, NuGet, Maven feeds	Artifactory / Nexus

PHASE 12 — INTERVIEW RAPID-FIRE COMPARISON TABLES

12.1 VM High Availability — Quick Decision Tree

Scenario	Solution	Why
2 VMs in same DC, survive hardware failure	Availability Set	Fault domains = separate racks
VMs survive entire datacentre failure	Availability Zones	3 physically separate DCs in one region
Auto-scale VMs based on load	VMSS	Identical VM pool with scaling rules
DR if entire region fails	Azure Site Recovery	Continuous replication to secondary region
Backup with 30-day retention	Azure Backup	Daily snapshots, file/VM restore

12.2 Storage Decision Matrix

Need	Azure Service
Store images, videos, any files (unstructured)	Azure Blob Storage
Shared network drive (SMB/NFS)	Azure File Shares
Async message queue between services	Azure Queue Storage
NoSQL key-value store for structured data	Azure Table Storage
Relational database — SQL Server compatible	Azure SQL Database
Open-source MySQL	Azure Database for MySQL
Global distributed NoSQL, multi-model	Azure Cosmos DB
Petabyte offline data transfer to Azure	Azure Data Box
Big data analytics / data warehouse	Azure Synapse Analytics
Apache Spark big data processing	Azure Databricks

12.3 Networking Cheat Sheet

Scenario	Service
Private connectivity between 2 VNets	VNet Peering
Secure browser-based RDP/SSH, no public IP needed	Azure Bastion
Distribute TCP/UDP traffic across VMs (same region)	Azure Load Balancer (Standard)
Route users to nearest region (DNS-based)	Azure Traffic Manager
Global CDN + WAF + HTTP LB	Azure Front Door
Encrypted tunnel to on-premises over internet	VPN Gateway
Private dedicated circuit to on-premises (no internet)	ExpressRoute
Custom routing through Firewall/NVA	User Defined Routes (UDR)
Diagnose NSG blocking traffic	Network Watcher — IP Flow Verify
Filter traffic at subnet/NIC level	Network Security Group (NSG)

Group VMs by role for NSG rules	Application Security Group (ASG)

12.4 Identity Quick Reference

Question	Answer
Which Entra ID licence for Conditional Access?	P1 minimum
Which licence for PIM (Privileged Identity Management)?	P2
What is SSPR and what does it require?	Self-Service Password Reset — requires P1 licence + authentication method registered
Owner vs Contributor?	Owner can manage access (RBAC). Contributor cannot.
Can RBAC Deny override Allow?	No — RBAC is additive Allow only. Use Azure Policy for Deny.
What is a Managed Identity?	Auto-managed service principal for Azure resources — no credentials, no secrets needed
What is Entra Domain Services vs Entra ID?	Entra ID = modern cloud identity (no domain join). Domain Services = managed legacy AD with LDAP/Kerberos/Group Policy

12.5 IaC / Governance Cheat Sheet

Question	Answer
ARM Template vs Bicep?	Same outcome. Bicep = cleaner syntax, transpiles to ARM JSON. Use Bicep for new projects.
ARM Template mode: Incremental vs Complete?	Incremental = add/update only. Complete = deletes resources not in template.
CanNotDelete Lock vs ReadOnly Lock?	CanNotDelete prevents deletion only. ReadOnly prevents ALL modifications including deletion.
Who can remove a resource lock?	Owner or User Access Administrator — regardless of other RBAC roles
Do tags inherit from resource group to resources?	No — tags are not inherited. Use Azure Policy to enforce inheritance.
Policy Deny vs Policy Audit?	Deny blocks creation. Audit allows creation but marks non-compliant.
Management Groups — max nesting depth?	6 levels below the root management group
Which scope inherits policy from above?	All levels — MG → Subscription → RG → Resource inherits from parent

12.6 Cost Management Services [AZ-900]

Tool	Purpose
Azure Pricing Calculator	Estimate monthly cost BEFORE deployment — configure services and get price estimate
TCO Calculator	Compare on-premises infrastructure cost vs Azure — Total Cost of Ownership

Azure Cost Management	Analyse and monitor ACTUAL spending in your Azure subscription after deployment
Budgets	Set spending thresholds with alerts — proactive notification before overspend
Azure Advisor	Free recommendations for cost, security, performance, reliability, and operational excellence

END OF NOTES — AZ-900 + AZ-104 COMBINED STUDY REFERENCE